
Algorithm 1 Gaussian speaker mechanism $\mathcal{M}_G$

Require: speaker embedding vector S from $\phi(x) = (S, C)$

Require: maximum L_2 perturbation, U

Require: privacy parameter, μ

$S' \leftarrow L_2 \text{clip}(S, U)$

$\sigma = \frac{U}{\mu}$

$\hat{S} = S' + \mathcal{N}(\sigma^2)$

Ensure: $(\hat{S}, C)$

Proof of the Gaussian Speaker Mechanism

Definition 1 (Speaker differential privacy). *A privacy mechanism $\mathcal{M}$ satisfies μ -speaker differential privacy if for all utterances x_1 spoken by S_1 and x_2 spoken by S_2 such that $\phi(x_1) = (S_1, C)$ and $\phi(x_2) = (S_2, C)$, then:*

$$T(\theta(\mathcal{M}(\phi(x_1))), \theta(\mathcal{M}(\phi(x_2)))) \geq G_\mu$$

Theorem 1. *The Gaussian speaker mechanism $\mathcal{M}_G$ satisfies μ -speaker differential privacy.*

Proof. We need to show that for all utterances x_1 spoken by S_1 and x_2 spoken by S_2 such that $\phi(x_1) = (S_1, C)$ and $\phi(x_2) = (S_2, C)$, it holds that:

$$T(\theta(\mathcal{M}_S(\phi(x_1))), \theta(\mathcal{M}_S(\phi(x_2)))) \geq G_\mu$$

Since the randomness comes only from the mechanism, we have:

$$\begin{aligned} & T(\theta(\mathcal{M}_S(\phi(x_1))), \theta(\mathcal{M}_S(\phi(x_2)))) \\ &= T(\mathcal{M}_S(\phi(x_1)), \mathcal{M}_S(\phi(x_2))) \\ &= T((\hat{S}_1, C_1), (\hat{S}_2, C_2)) && (\text{defn } \mathcal{M}_S) \\ &= T(\hat{S}_1, \hat{S}_2) && (C_1 = C_2) \\ &= T(\mathcal{N}(S'_1, \sigma^2), \mathcal{N}(S'_2, \sigma^2)) && (\text{defn } \mathcal{M}_S) \\ &= G_{(S'_1 - S'_2)/\sigma} && (\text{defn of } T \text{ and } G) \\ &\geq G_{U/\sigma} && (\text{sensitivity by clipping}) \\ &= G_\mu && (\text{defn of } \mathcal{M}_S) \end{aligned}$$

□